

Ex06. Introduction to Signals and systems

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1. Plot 3*3 Charts with Function sinc()

This assignment involves creating and discussing 3*3 signal charts for the sinc function. For the numerical computation, the discrete time index n from -40 to 40 with an increment of 1. The frequency domain ω is sampled at intervals of 0.1 (from -2π to 2π), and the constant ω_c is set to 2. Change the limited length setting and compare the results output via the "phase()" and "abs()" functions.

```
n = -40:1:40;
n10 = -10:1:10;
omega = -2*pi:0.1:2*pi;
omega_c = 2;

h = @(n) (omega_c / pi) .* sinc((omega_c .* n) ./ pi);
H = @(omega) (abs(omega) < omega_c);

H10 = zeros(size(omega));
for i = n10
    H10 = H10 + h(i) .* exp(-1j .* omega .* i);
end

H40 = zeros(size(omega));
for i = n
    H40 = H40 + h(i) .* exp(-1j .* omega .* i);
end

phase40 = angle(H40);
phase40 = (phase40);
phase10 = angle(H10);
phase10 = (phase10);
figure('Position', [100, 100, 1000, 1000]);

subplot(3, 3, 1);
plot(n, h(n), 'LineWidth', 2);
title('h[n]');
xlabel('n');
ylabel('Amplitude');
grid on;

subplot(3, 3, 2);
plot(omega, phase(H(omega)), 'LineWidth', 2);
title('phase(H(\omega))');
xlabel('Frequency');
ylabel('Phase [rad]');
grid on;
```

```

subplot(3, 3, 3);
plot(omega, abs(H(omega)), 'LineWidth', 2);
title('abs(H(\omega))');
xlabel("Frequency");
ylabel('Amplitude');
grid on;

subplot(3, 3, 4);
plot(n, h(n), 'LineWidth', 2);
title('h_{40}(n)');
xlabel("n");
ylabel('Amplitude');
grid on;

subplot(3, 3, 5);
plot(omega, phase40, 'LineWidth', 2);
title('phase(H_{40}(\omega))');
xlabel("Frequency");
ylabel('Phase [rad]');
grid on;

subplot(3, 3, 6);
plot(omega, abs(H40), 'LineWidth', 2);
title('abs(H_{40}(\omega))');
xlabel("Frequency");
ylabel('Amplitude');
grid on;

subplot(3, 3, 7);
plot(n10, h(n10), 'LineWidth', 2);
title('h_{10}(n)');
xlabel("n");
ylabel('Amplitude');
grid on;

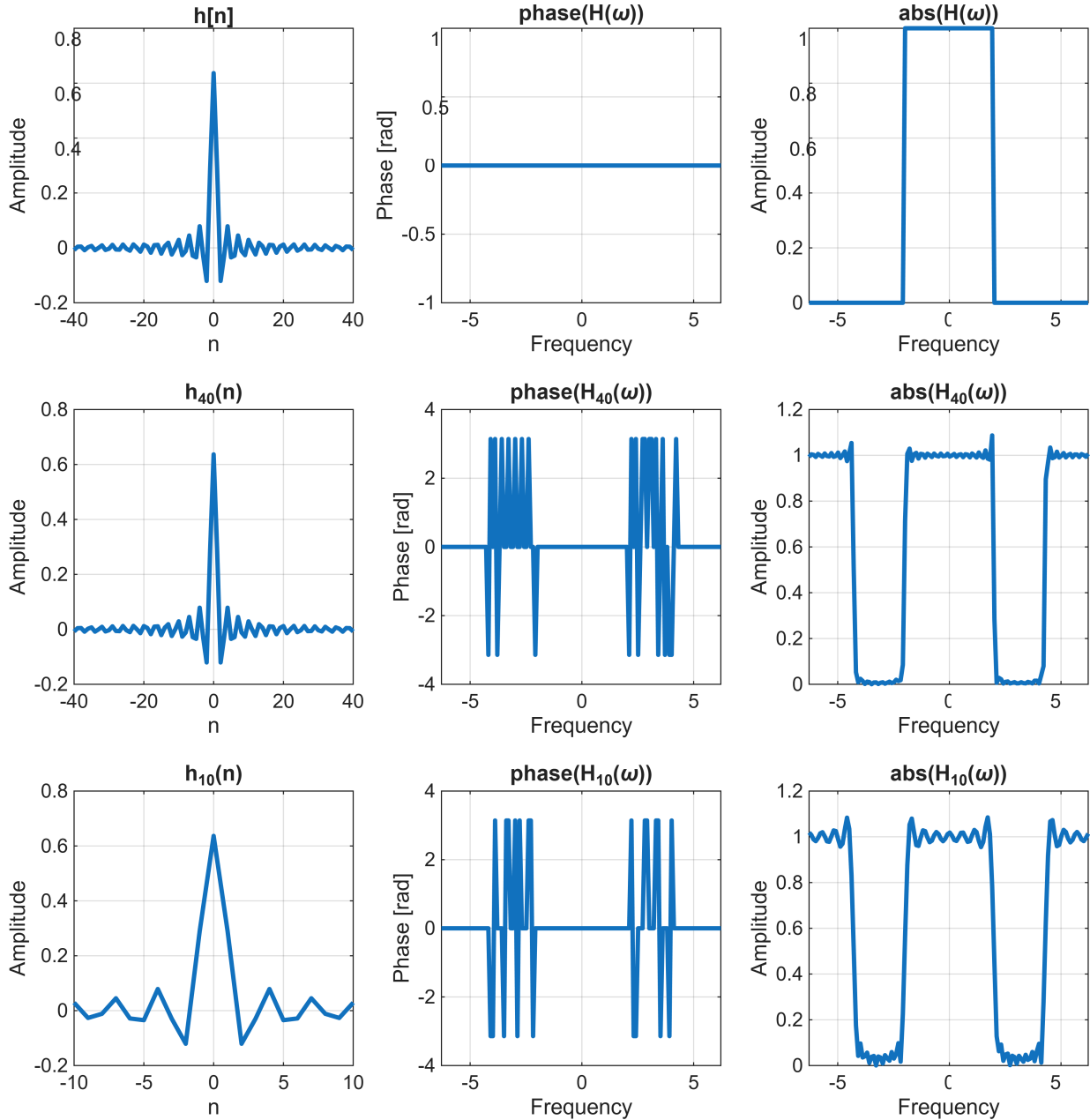
subplot(3, 3, 8);
plot(omega, phase10, 'LineWidth', 2);
title('phase(H_{10}(\omega))');
xlabel("Frequency");
ylabel('Phase [rad]');
grid on;

subplot(3, 3, 9);
plot(omega, abs(H10), 'LineWidth', 2);
title('abs(H_{10}(\omega))');
xlabel("Frequency");
ylabel('Amplitude');
grid on;

sgtitle('Sinc Signal Analysis');

```

Sinc Signal Analysis



1.2. Discussion

The figures plotted above show several remarkable trends. First of all, the graphs in the first row describe the ideal behavior of each function ($h[n]$, $H(\omega)$), where $\text{abs}(H(\omega))$ exhibits a perfect rectangular shape and $\text{phase}(H(\omega))$ is constantly 0 across the entire frequency range.

The figures located below describe the behavior when the impulse response $h[n]$ is truncated to a finite length. Regarding $\text{phase}(H(\omega))$ and $\text{abs}(H(\omega))$, after applying the DTFT to the finite-length segments, ripples (Gibbs phenomenon) begin to appear near the cutoff frequencies. This tendency is particularly pronounced

in $\text{abs}(H(\omega))$; the shorter the limited length of the impulse response, the larger the ripples become, failing to maintain the perfect rectangular shape.

2. Math Problem

Below is the snapshot of my answer in fill-blank math problem.

$$\begin{aligned} & \frac{1}{2\pi} \int_{-W_c}^{W_c} e^{j\omega t} d\omega // \frac{1}{j2\pi t} [e^{j\omega t}]_{-W_c}^{W_c} // \\ & \frac{e^{jW_c t} - e^{-jW_c t}}{j2\pi t} // \frac{\sin(W_c t)}{\pi t} // \frac{W_c}{\pi} \cdot \frac{\sin(W_c t)}{W_c t} // \\ & \frac{W_c}{\pi} \cdot \text{sinc}\left(\frac{W_c t}{\pi}\right) // \frac{W_c}{\pi} \cdot \frac{\sin(W_c t)}{W_c t} // \\ & \frac{W_c}{\pi} \cdot \text{sinc}\left(\frac{W_c t}{\pi}\right) // \end{aligned}$$